

CSC343H5 Introduction to Databases
Sprint 4: Normalisation

Released: Friday, July 10th **Due:** Sunday, July 26th at 5PM ET
Resubmission: one, within one week of grade release (see §6)

1. Overview

In earlier sprints you designed the Research Network schema from scratch. Here you work in reverse: you are given a deliberately *denormalised* table, **research_flat**, that collapses grants, projects, researchers, and their relationships into one relation. Your task is to analyse its functional dependencies, compute a minimal cover, and decompose it into BCNF. When you finish, compare your result to the grant, project, and researcher tables from your Sprint 2 DDL, they should look structurally similar.

2. The Flat Table

research_flat has the composite key (**grant_code**, **project_id**, **researcher_id**): one row means a researcher works on a project funded by a grant.

Attributes: grant_code, grant_title, funding_body, funding_sector, project_id, project_title, budget, researcher_id, researcher_name, institution, role_on_project.

Sample rows

grant	funder	sector	project	researcher	role
NSERC-4471	NSERC	federal	EcoForecast	Amara Osei	PI
NSERC-4471	NSERC	federal	EcoForecast	Priya Sharma	Postdoc
NSERC-8810	NSERC	federal	NetSec-QoS	Lena Hartmann	PI
ORF-0117	ORF	provincial	NetSec-QoS	Lena Hartmann	PI
CIHR-9934	CIHR	federal	OmicsIntegrator	Mia Chen	PI
CIHR-9934	CIHR	federal	OmicsIntegrator	Yuki Tanaka	Grad

(Titles and budgets are omitted from the sample for space; they repeat in the same pattern.)

Anomalies this table produces.

- **Update:** grant_title for NSERC-4471 is stored on every row; renaming means editing all copies.
- **Insertion:** a grant with no project or researcher yet cannot be stored, part of the key would be missing.
- **Deletion:** removing the last researcher on OmicsIntegrator also erases the fact that CIHR funds it.
- **Redundancy:** funding_sector repeats on every row of a funder, though it is a fact about the funder.

3. Functional Dependencies

The complete set of non-trivial FDs that hold in **research_flat** is given below. Use exactly these, do not add or modify them. FD1 to FD3 say “an id determines its own facts”; FD4 is a transitive dependency (**funding_sector** is a property of the funder, not the grant); FD5 says a researcher’s role is fixed by the specific grant, project, and researcher together.

- **FD1.** $\text{grant_code} \rightarrow \text{grant_title}, \text{funding_body}$
- **FD2.** $\text{project_id} \rightarrow \text{project_title}, \text{budget}$
- **FD3.** $\text{researcher_id} \rightarrow \text{researcher_name}, \text{institution}$
- **FD4.** $\text{funding_body} \rightarrow \text{funding_sector}$
- **FD5.** $\text{grant_code}, \text{project_id}, \text{researcher_id} \rightarrow \text{role_on_project}$

4. Tasks

Task 1: Closure & Candidate Key (20%)

1. Compute $\{\text{grant_code}, \text{project_id}, \text{researcher_id}\}^+$ from the FDs in §3, showing each step. Confirm it yields every attribute.
2. Confirm the composite key is the only candidate key (or identify any others), with a one-sentence justification.
3. List the prime and non-prime attributes. You will use this in Task 4.

Task 2: Minimal Cover (25%)

Reduce the FD set from §3 to a minimal cover, showing all three steps:

1. Split every right-hand side to a single attribute.
2. Remove redundant left-hand-side attributes (show the closure test for each multi-attribute FD).
3. Remove redundant FDs (show the closure test for each).

State the resulting minimal cover.

Task 3: BCNF Decomposition (35%)

Decompose **research_flat** into BCNF using your minimal cover. For **each** step, state:

- the violating FD $X \rightarrow Y$ (why X is not a superkey of that relation),
- the two resulting relations and each relation’s primary key, and
- the **lossless check**: the shared attributes must be a key of one side.

Then give your final BCNF schema in schema notation (**table**(**pk**, col, ...), primary keys underlined) and confirm each relation is in BCNF (every non-trivial FD has a superkey left-hand side).

Task 4: 3NF Check & Justification (20%)

1. **Dependency preservation:** check whether every FD in your minimal cover can be verified inside a single relation (no join). State whether your BCNF schema is dependency-preserving; if not, give the 3NF synthesis result and say which you would use and why.
2. **Anomaly example:** pick *one* anomaly type and give a concrete example using the sample rows in §2 (which rows, what goes wrong).

5. Submission & Policies

Submit to the course portal by **July 26th, 5 PM ET**: `s4.tex` (source) and `s4.pdf` (compiled). Put your full name, student number, and UTORid in `s4.tex`.

Resubmission. One resubmission within one week of grade release. The FD set and minimal cover are graded on first submission only; on resubmission you may revise the decomposition and justification (include a short changelog).

AI usage. AI use is permitted but must be declared. AI tools often get closures and minimal-cover steps wrong, you are graded on the correctness and justification of *your* work, so verify anything a tool suggests.

6. Rubric

Criterion	Marks
Task 1: closure shown step by step; candidate key confirmed; prime/non-prime attributes listed.	20
Task 2: three minimal-cover steps shown with closure tests; correct result.	25
Task 3: each step (violation, relations, keys) and lossless check shown; final BCNF schema correct and verified.	35
Task 4: dependency-preservation analysis (3NF synthesis if needed); concrete anomaly example.	20
Total	100